Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Development of AI Predictive Maintenance Smart Factory Platform Technology Based on Integrated Sensing of Power Quality, Leakage, and Heat 전력품질·누설·열 복합 센싱 기반 AI 예지보전 스마트팩토리 플랫폼 기술 개발			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical·Electronics		Information & Communication	
	Medium Category	Small Category	Medium Category	Small Category
	Heavy Electrical Equipment	Transmission/ Distribution, Protection and Monitoring	Software	Software solution
Concept of Proposed Technology	This technology aims to develop an intelligent service that monitors high-power facilities (motors, compressors, etc.) in smart factories. It integrates and analyzes 3-phase power quality sensors, leakage current, and distribution panel thermal information to predict failures (Predictive Maintenance) and respond to hazardous situations in real-time. Unlike existing technologies focused on single sensors (vibration, noise, or temperature), this technology interprets the correlation between electrical causes (unbalance, reverse phase, harmonics, leakage, low power factor) and mechanical/thermal anomalies using AI, offering the following differentiators:			

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	● Early detection of "electrical anomaly signs" before facility shutdown ● Immediate applicability to standard power infrastructure in smart factories and overseas plants ● Global export-oriented structure considering North American/European safety and power standards
Rationale for Joint R&D Project	Securing original technology through international joint R&D ● Strengthening technical reliability for global market entry ● Field verification and demand-based development through collaboration with Korean SMEs ● Establishing an R&D structure that leads directly to commercialization and export
Key Words	1. 3-phase power-based predictive maintenance 2. Integrated diagnosis of power quality/leakage/heat 3. Early detection of electrical anomalies 4. Explainable AI (XAI) based facility analysis 5. Export-oriented smart factory predictive maintenance platform

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (7)
R&D Objectives	**Qualitative Objectives:** ● Securing technology for diagnosing preceding signs of facility failure using 3-phase power data. ● Developing an Explainable AI (XAI) predictive maintenance model fusing electrical, thermal, and leakage information. ● Implementing real-time hazard response services applicable to smart factory sites. ● Securing global technical competitiveness targeting domestic verification and the Canada/North American markets. **Quantitative Objectives:** ● Data collection success rate: ≥ 90%. ● Facility failure classification accuracy: ≥ 90%. ● Lead time for early detection of major failure types (unbalance, insulation degradation, overload, etc.): 7 to 30 days. ● Reduction of unplanned downtime in smart factories: ≥ 30%.			
R&D Target Specifications	**(1) Sensor/Measurement Specifications:** ● Real-time measurement of 3-phase voltage, current, power (P, Q, S), power factor, and energy consumption. ● Voltage and current unbalance rates, symmetric components (positive, negative, and zero sequence). ● Harmonics (THD, major orders) and detection of start/stop events. ● Resistive leakage current (Ir) and zero-sequence current (I0). ● Multi-point temperature sensing in distribution panels. **(2) Data/Analysis Specifications:** ● High-resolution data collection (≥ 6.4kHz waveform or event capture). ● Automatic generation of normal state baselines for each facility. ● AI models for anomaly detection and failure type classification. ● Application of multi-task models adaptable to various types of facilities. ● Explainable analysis results providing guides for the cause, impact, and countermeasures of anomalies. **(3) Service/System Specifications:** ● Real-time alarms and multi-stage risk notifications. ● Web-based integrated smart factory dashboard. ● Cloud/Edge hybrid architecture. ● Architecture compatible with both North American and domestic (Korean) power standards.			
R&D Plan	**1. Research Approach** This project is conducted as convergence research combining Electrical Engineering, Data Science, and Industrial AI. ● **Domestic Research Team:** ■ Definition of industrial site requirements. ■ Demonstration of power sensors and smart factory applications. ■ Design of service platforms and commercialization models. ● **Collaboration with University of Toronto:** ■ Theoretical research on anomaly detection based on power signals. ■ Advanced analysis models for harmonics, unbalance, and leakage patterns. ■ Generalization of AI models and verification of overseas applicability. **2. Key R&D Tasks**			

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	● Development of facility status diagnosis algorithms based on 3-phase power: ■ Analysis of unbalance, negative sequence, harmonics, and starting patterns. ● Integrated risk prediction model for leakage current and temperature: ■ Early warning systems for fire and insulation degradation. ● Development of AI predictive maintenance models: ■ Anomaly detection, failure type classification, and Remaining Useful Life (RUL) estimation. ● Technology for generating explainable diagnosis reports: ■ Providing results that are easily understandable by on-site maintenance personnel. ● Smart factory demonstration and validation for overseas expansion: ■ Verification through Korean factories and Canadian collaborative testbeds. **3. Methodology** ● Statistics-based anomaly detection (EWMA, change-point analysis). ● Machine learning and deep learning-based unsupervised and semi-supervised learning. ● Event-driven data analysis (focusing on starting and anomaly waveforms). ● Correlation analysis of electrical, thermal, and leakage data.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	● Development of AI predictive maintenance models and analysis algorithms based on 3-phase power data ● Establishing theoretical interpretation models for power quality, leakage, and thermal data ● Designing anomaly pattern analysis and prediction logic by failure type ● Researching Explainable AI (XAI) based diagnostic algorithms ● * Verifying generalization performance and global applicability of AI models
Expected R&R of Korean SME Partner	● Development of 3-phase power quality sensors and measurement systems ● Construction of data collection infrastructure for smart factory sites ● Facility data collection and quality management in verification environments ● Field application and performance verification of predictive maintenance technology ● Commercialization and productization of research results

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**1. Technical Outcomes:** ● Securing original predictive maintenance technology for high-power facilities by fusing 3-phase power quality (unbalance, reverse phase, harmonics), leakage current, and distribution panel temperature. ● Implementing early detection technology for electrical anomaly signs prior to failure, surpassing single-phase/single-sensor-based predictive maintenance technologies. ● Developing status diagnosis algorithms for major facilities such as motors and compressors through the analysis of facility start/stop and operation patterns. ● Securing explainable predictive maintenance algorithms that combine electrical engineering theory and AI analysis. ● Implementing real-time diagnosis, alarm, and hazard response technologies applicable to smart factory sites. **2. Commercialization Potential:** ● Establishing a modular predictive maintenance platform architecture consisting of sensors, data collection, analysis, and services. ● Securing technology readiness levels (TRL) that allow for immediate commercialization in the form of products and services upon project completion. ● Securing technical reliability based on field verification through demonstrations with Korean SMEs. ● Possibility of business expansion into various industrial fields such as smart factory maintenance, facility management, and safety management. ● Potential for expansion into export-oriented solutions for North American and global smart factory markets after leading the domestic market. ● Securing global technical competitiveness and brand credibility utilizing the outcomes of international joint research. **3. Industrial & Economic Impact:** ● Improving productivity at manufacturing sites by reducing unplanned downtime caused by facility failure. ● Creating operating cost reduction effects through maintenance cost savings and the extension of facility lifespan. ● Preventing industrial accidents and improving safety levels through advance detection of electrical anomalies and fire risks. ● Lowering barriers for SMEs to adopt smart factories and accelerating digital transformation. ● Enhancing domestic technical independence by replacing foreign predictive maintenance solutions. ● Expanding global smart factory exports centered on SMEs and strengthening national industrial competitiveness.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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